Project : Scene-Aware Multimodal Pedestrian Trajectory Prediction

T2IA -Deep Learning and Genrative AI

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https://github.com/ZhongtianF/Trajectory-Prediction

Abstract

Pedestrian trajectory prediction aims to forecast future human motion from previously observed positions. This task is challenging because human motion is uncertain and multiple future trajectories may be possible for the same observed path.

In this project, three trajectory prediction approaches were implemented and evaluated on the ETH/UCY dataset. First, a Constant Velocity baseline was used as a simple reference model. Second, a deterministic LSTM encoder–decoder model was trained to learn motion patterns from trajectory sequences. Finally, a GAN-based model was implemented to generate multiple possible future trajectories.

Model performance was evaluated using Average Displacement Error (ADE) and Final Displacement Error (FDE). For the stochastic model, Best-of-K sampling was applied to evaluate multimodal prediction quality.

Experimental results show that the deterministic LSTM does not outperform the Constant Velocity baseline, while the GAN model achieves the best performance under Best-of-20 evaluation. These results highlight the importance of multimodal prediction for modeling uncertain pedestrian behavior.

I. Introduction

Pedestrian trajectory prediction aims to estimate future pedestrian positions based on a short sequence of observed motion. This problem is important for applications such as autonomous driving, mobile robots, and crowd analysis, where anticipating human movement is necessary for safe navigation.

A major difficulty of trajectory prediction is that pedestrian motion is inherently uncertain. For the same observed trajectory, multiple future paths may exist. For example, a pedestrian approaching an intersection may continue forward or change direction. Therefore, predicting only a single trajectory may not fully represent possible future behaviors.

In this project, deterministic and stochastic approaches for trajectory prediction are studied using the ETH/UCY pedestrian dataset. First, a Constant Velocity baseline is implemented as a simple reference model. Then, a deterministic LSTM encoder–decoder model is trained to predict future trajectories from past motion. Finally, a Generative Adversarial Network (GAN) is implemented to generate multiple possible future trajectories by sampling latent noise.

The models are evaluated using standard trajectory prediction metrics including Average Displacement Error (ADE) and Final Displacement Error (FDE). In addition to quantitative evaluation, qualitative visualizations on scene images are used to analyze prediction behavior.

The rest of this report is organized as follows. Section 2 describes the dataset and preprocessing steps. Section 3 presents the implemented models. Section 4 explains the experimental setup. Section 5 reports the results and analysis. Finally, Section 6 concludes the report.

II. Problem Formulation

Pedestrian trajectory prediction can be formulated as a **sequence forecasting problem**. Given a sequence of observed pedestrian positions, the objective is to predict the future trajectory over a predefined prediction horizon.

A pedestrian trajectory can be represented as a sequence of 2D spatial coordinates over time:

$$T = \{(x_1, y_1), (x_2, y_2), \ldots, (x_t, y_t)\}$$

In this project, each trajectory is divided into two parts:

Observed trajectory

$$T_{obs} = \{(x_1, y_1), \ldots, (x_8, y_8)\}$$

Future trajectory

$$T_{pred} = \{(x_9, y_9), \ldots, (x_{20}, y_{20})\}$$

The goal of the model is to learn a function that predicts the future trajectory based on the observed trajectory:

$$T_{pred} = f(T_{obs})$$

Following the standard **ETH/UCY evaluation setting**, the model observes **8 time steps** and predicts the **next 12 future positions**.

Evaluation Metrics

Model performance is evaluated using two widely used trajectory prediction metrics.

Average Displacement Error (ADE) measures the average Euclidean distance between predicted positions and ground-truth positions across all predicted time steps.

Final Displacement Error (FDE) measures the Euclidean distance between the predicted final position and the ground-truth final position.

For stochastic models that generate multiple possible trajectories, **Best-of-K evaluation** is used. In this setting, multiple predictions are generated and the prediction with the smallest error relative to the ground truth is selected.

III. Dataset and Preprocessing

The models implemented in this project were trained and evaluated using the **ETH/UCY pedestrian trajectory dataset**, which is a widely used benchmark for human trajectory prediction. The dataset contains real-world pedestrian movements recorded in outdoor environments such as university campuses and public walkways. It is commonly used to evaluate trajectory forecasting models because it contains complex pedestrian behaviors and diverse motion patterns.

The dataset consists of **five different scenes**:

- ETH
- HOTEL
- UNIVERSITY
- ZARA1
- ZARA2

Each scene contains multiple pedestrian trajectories extracted from video sequences. A trajectory is represented as a sequence of **2D spatial coordinates** describing the position of a pedestrian at each time step.

Data Format

In this project, the trajectory data are stored in **SQLite databases**. Each record corresponds to the state of a pedestrian at a specific frame. The main attributes stored in the database include:

- **pos_x** : x-coordinate of the pedestrian position
- **pos_y** : y-coordinate of the pedestrian position
- **pos_x_delta** : displacement in the x direction between frames
- **pos_y_delta** : displacement in the y direction between frames
- **ped_id** : unique pedestrian identifier
- **frame_num** : frame index
- **data_id** : trajectory sequence identifier

These attributes allow the reconstruction of continuous pedestrian trajectories over time.

Trajectory Segmentation

To prepare the data for trajectory prediction, continuous trajectories are segmented into shorter sequences following the standard **ETH/UCY evaluation protocol**.

Each trajectory segment contains **20 time steps**, divided into:

- **8 observed positions** used as model input
- **12 future positions** used as prediction targets

This **8→12 setting** is commonly used in trajectory prediction research and allows fair comparison between different models.

Dataset Statistics

After preprocessing and segmentation, the dataset contains the following number of trajectory samples for each scene:

Scene	Number of trajectory sequences
ETH	364
HOTEL	1197
UNIVERSITY	10039
ZARA1	2356
ZARA2	5910

The **University** scene contains the largest number of trajectories because the video sequence is longer and includes a higher pedestrian density.

To improve training stability, trajectory coordinates were **normalized using scene-specific statistics**, including mean and standard deviation. This normalization ensures that trajectories from different scenes are represented at a comparable scale during model training.

IV. Model Description

This section describes the trajectory prediction models implemented in this project. Three different approaches were explored: a simple physics-based baseline, a deterministic neural network model, and a stochastic generative model designed to capture multimodal future trajectories.

4.1 Constant Velocity Baseline

The Constant Velocity (CV) model serves as a simple baseline for trajectory prediction. This approach assumes that a pedestrian will continue moving with the same velocity observed in the last time steps.

Given the observed trajectory:

$$X_{obs} = \{(x_1, y_1), \ldots, (x_{T_{obs}}, y_{T_{obs}})\}$$

the velocity is estimated using the displacement between the last observed positions:

$$v = (x_{T_{obs}} - x_{T_{obs}-1}, y_{T_{obs}} - y_{T_{obs}-1})$$

Future positions are then predicted by extrapolating this velocity:

$$\hat{p}_{t+1} = p_t + v$$

for the entire prediction horizon.

Although this model is extremely simple, it provides an important reference point. More complex models should ideally outperform this baseline in order to demonstrate meaningful learning.

4.2 Deterministic Trajectory Prediction with LSTM

To learn motion patterns from data, a neural network model based on **Long Short-Term Memory (LSTM)** was implemented. LSTM networks are well suited for sequence modeling tasks because they can capture temporal dependencies over time.

The deterministic LSTM model follows an **encoder – decoder architecture**.

First, the encoder processes the observed trajectory sequence:

$$X_{obs} = \{(x_1, y_1), \ldots, (x_{T_{obs}}, y_{T_{obs}})\}$$

Each coordinate pair is embedded and fed into an LSTM layer. The final hidden state of the encoder represents the encoded motion context of the pedestrian.

The decoder then generates the predicted trajectory step by step. At each time step, the decoder predicts the next position based on the previous hidden state and the previously predicted position.

Formally, the model can be written as:

$$h = Encoder(X_{obs})$$

$$\hat{X}_{pred} = Decoder(h)$$

where h is the encoded hidden representation of the observed trajectory.

The model outputs a single predicted trajectory of length 12 time steps.

Figure 1: Deterministic trajectory prediction using the LSTM model. Blue indicates the observed trajectory, green shows the ground-truth future path, and red represents the predicted trajectory generated by the deterministic model.

4.3 GAN-Based Multimodal Trajectory Prediction

While deterministic models generate a single future trajectory, real-world pedestrian motion is inherently uncertain. A pedestrian approaching an intersection may have several possible future paths. To model this uncertainty, a **Generative Adversarial Network (GAN)** was implemented to generate multiple plausible trajectories.

The GAN architecture consists of two neural networks:

Generator G

Discriminator D

Generator

The generator produces predicted trajectories conditioned on the observed motion.

The generator receives three inputs:

Observed trajectory

$$X_{obs}$$

Random latent noise vector

$$z \sim N(0,1)$$

Scene features extracted from the scene image.

These inputs are combined to generate a future trajectory:

$$\hat{X}_{pred} = G(X_{obs}, z, S)$$

where S represents the scene feature vector.

Because the noise vector z changes at each sampling step, the generator can produce **different plausible future trajectories**.

Discriminator

The discriminator is trained to distinguish between real trajectories and generated trajectories.

It receives the full trajectory sequence:

$$[X_{obs}, X_{pred}]$$

and outputs a probability indicating whether the trajectory is real or generated.

The generator is trained to fool the discriminator, while the discriminator is trained to correctly classify real and generated trajectories.

This adversarial training encourages the generator to produce realistic trajectory distributions.

4.4 Best-of-K Sampling

To evaluate stochastic trajectory prediction models, the **Best-of-K sampling strategy** was implemented.

For each observed trajectory, the generator produces multiple candidate future trajectories:

$$\hat{X}_{pred}^{(1)}, \hat{X}_{pred}^{(2)}, \ldots, \hat{X}_{pred}^{(K)}$$

Each prediction is compared with the ground truth trajectory. The prediction with the smallest error is selected:

$$ADE_K = \min_k \; ADE(\hat{X}_{pred}^{(k)}, X_{pred})$$

$$FDE_K = \min_k \; FDE(\hat{X}_{pred}^{(k)}, X_{pred})$$

In this project, **K = 20** samples were generated for each observed trajectory.

This evaluation reflects the model's ability to generate at least one plausible future trajectory among several possible outcomes.

Figure 2: Multimodal trajectory prediction generated by the GAN model. Multiple sampled future trajectories are shown in light red, while the best prediction among the samples is highlighted in orange.

V. Experimental Setup

This section describes the training configuration, evaluation protocol, and implementation details used in the experiments.

Training Configuration

All models were implemented using **PyTorch** and trained on the ETH/UCY trajectory dataset described in the previous section. Trajectories were normalized using scene-specific mean and standard deviation statistics before being used as model inputs.

For both the LSTM and GAN models, the observation length was set to **8 time steps**, while the prediction horizon was **12 time steps**, following the standard evaluation protocol used in trajectory prediction literature.

The models were trained using the **Adam optimizer**, which is widely used for training neural networks due to its adaptive learning rate properties. Training was performed for **20 epochs**, and the model with the best validation performance was saved for evaluation.

The GAN model was trained using an adversarial learning framework consisting of a generator and a discriminator. In addition to the adversarial loss, a trajectory reconstruction loss was also used to encourage the generator to produce realistic future trajectories.

Scene context information was incorporated by extracting features from the scene images using a pre-trained **ResNet18**backbone. These scene features were used as additional conditioning inputs to the trajectory generator.

Evaluation Protocol

Model performance was evaluated using two standard trajectory prediction metrics:

Average Displacement Error (ADE)

Final Displacement Error (FDE)

ADE measures the average Euclidean distance between predicted positions and ground-truth positions over all predicted time steps, while FDE measures the Euclidean distance between the predicted final position and the ground-truth final position.

For stochastic models such as GANs, the **Best-of-K evaluation protocol** was adopted. In this approach, the model generates multiple candidate future trajectories for each observed trajectory. The prediction with the smallest error relative to the ground truth is selected for evaluation.

In this project, **K = 20** future trajectories were sampled for each observed trajectory when computing Best-of-K performance.

VI. Experimental Results

Table 1 — Overall Model Performance

Model	ADE	FDE
Constant Velocity	0.5493	1.2102
LSTM	0.6276	1.2192
GAN (single sample)	0.6576	1.2749
GAN (Best-of-20)	**0.5422**	**1.0610**

Overall Performance

The quantitative results of the different models are summarized in Table 1.

The Constant Velocity baseline provides a simple reference for trajectory prediction. Despite its simplicity, it performs reasonably well because short-term pedestrian motion often follows approximately linear dynamics.

The deterministic LSTM model did not outperform the Constant Velocity baseline in terms of overall ADE and FDE. This result suggests that predicting a single deterministic trajectory may not be sufficient to capture the uncertainty present in pedestrian motion.

The GAN model produces slightly worse results when only a single sampled trajectory is evaluated. However, when using the **Best-of-20 evaluation strategy**, the GAN achieves the best overall performance. In particular, the Best-of-20 model obtains the lowest ADE and FDE among all evaluated models.

These results highlight the importance of modeling trajectory prediction as a **multimodal problem**, where multiple plausible future trajectories may exist for the same observed motion.

Table 2 — Scene-Level Performance

Scene	CV ADE	LSTM ADE	GAN ADE	GAN Best-of-20 ADE
ETH	0.7709	0.8674	1.0940	0.8857
HOTEL	0.3534	0.4911	0.3765	0.2351
UNIVERSITY	0.6795	0.6932	0.7285	0.5988
ZARA1	0.4123	0.4637	0.5520	0.4603
ZARA2	0.4086	0.5944	0.6090	0.5198

Scene-Level Analysis

Performance across individual scenes shows that the trajectory prediction difficulty varies depending on the environment.

The HOTEL scene produces the lowest prediction errors for all models. This scene contains relatively simple pedestrian motion patterns with fewer complex interactions.

In contrast, the ETH and UNIVERSITY scenes are more challenging due to higher pedestrian density and more complex motion patterns.

The GAN model demonstrates clear advantages when evaluated with Best-of-K sampling, particularly in scenes with more diverse motion patterns.

VII. Failure Analysis

Although the GAN-based trajectory prediction model improves performance under Best-of-K evaluation, the model still exhibits several failure cases. These failures typically occur in scenarios involving complex human motion patterns or sudden changes in direction.

Figure 3: Example of a failure case in multimodal trajectory prediction. The predicted trajectories diverge from the ground truth due to complex motion patterns and uncertainty in pedestrian behavior.

Uncertainty in Human Motion

One major source of prediction error is the inherent uncertainty in human behavior. Pedestrians may suddenly change direction, stop, or

interact with other people in ways that are difficult to infer from past motion alone.

For example, when a pedestrian approaches an intersection or crowded area, multiple possible future trajectories become plausible. Even a multimodal model may fail if the true future trajectory deviates significantly from the dominant patterns learned during training.

Lack of Explicit Social Interaction Modeling

Another limitation of the implemented model is the absence of explicit modeling of interactions between pedestrians. Although multiple pedestrians may appear in the scene images, the current model predicts the trajectory of only one target pedestrian at a time.

As a result, the model does not explicitly account for collision avoidance, group behavior, or other social interactions that influence pedestrian motion. In crowded scenes, this limitation may lead to unrealistic or inaccurate predictions.

Future work could address this issue by incorporating social interaction mechanisms such as social pooling, graph neural networks, or attention-based models.

Limited Scene Understanding

Although scene features are extracted using a pre-trained ResNet18 model, the current implementation uses only global scene features rather than detailed local spatial information.

Consequently, the model may fail to properly interpret scene constraints such as walkable areas, obstacles, or pathways. A more advanced approach could integrate spatial attention mechanisms or semantic scene segmentation to better capture environmental structure.

VIII. Conclusion

This project investigated pedestrian trajectory prediction using deterministic and stochastic models on the ETH/UCY benchmark dataset. A complete prediction pipeline was implemented, including dataset preprocessing, trajectory modeling, evaluation, and visualization.

First, a Constant Velocity baseline was implemented to provide a simple reference for trajectory forecasting. A deterministic LSTM model was then trained to predict a single future trajectory based on the observed motion. Finally, a GAN-based model was introduced to generate multiple plausible future trajectories through stochastic sampling.

Experimental results show that deterministic trajectory prediction does not always outperform simple motion extrapolation baselines. In contrast, the GAN-based model demonstrates improved performance when evaluated using the Best-of-K sampling strategy. By generating multiple candidate trajectories, the stochastic model is able to capture the multimodal nature of pedestrian motion.

Qualitative visualizations further highlight the difference between deterministic and multimodal predictions. While deterministic models generate only one possible future path, the GAN model produces a diverse set of candidate trajectories that better reflect the uncertainty present in real-world human motion.

Despite these improvements, several limitations remain. The current implementation does not explicitly model social interactions between pedestrians and relies on global scene features rather than detailed spatial context. Addressing these limitations could further improve prediction accuracy in complex environments.

Overall, the results demonstrate that incorporating stochastic modeling and multimodal sampling is an effective approach for improving pedestrian trajectory prediction.